Vol 13 Chapter 6 — Biochemistry Bridge

Keeper (author pass — deep math/physics revision)

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Chapter 6 — Biochemistry Bridge

Level 1 — one sentence

Biochemistry — the chemistry of living systems including enzyme catalysis, metabolic pathways, ATP energy currency, photosynthesis, respiration — is the substrate K-type rearrangements of biological macromolecules under living-system boundary conditions, bridging Vol 12 chemistry to Vol 13 biology.

Level 2 — graduate-physicist precision

6.1 Enzyme catalysis

Enzymes lower activation energy by factors of 10^5 - 10^{17} (Vol 12 Ch 5).

Michaelis-Menten kinetics:

$$v = \frac{V_{\max}[S]}{K_M + [S]}$$

with $V_{\max}$ maximum rate, K_M Michaelis constant.

Active site provides substrate-K-type-optimal saddle path (Vol 12 Ch 5 transition state).

6.2 ATP energy currency

Adenosine triphosphate. Hydrolysis: ATP → ADP + Pi + 30.5 kJ/mol (at standard conditions; ~50 kJ/mol in vivo).

Universal energy currency across all known life. ~10^21 ATP molecules / human / day.

Synthesized by ATP synthase (mitochondria) via chemiosmotic proton gradient (Mitchell 1961, Nobel 1978).

6.3 Metabolic pathways

- Glycolysis: glucose → 2 pyruvate + 2 ATP + 2 NADH (~10 steps)
- Citric acid cycle: pyruvate → CO_2 + ATP + NADH + $FADH_2$ (~10 steps)
- Oxidative phosphorylation: NADH + O_2 → H_2O + ~30 ATP (chemiosmosis)
- Photosynthesis (light → chemical energy): light + H_2O + CO_2 → glucose + O_2

Net glucose oxidation: $C_6H_{12}O_6 + 6O_2 \to 6CO_2 + 6H_2O$, $\Delta G \approx -2870$ kJ/mol → ~30-32 ATP/glucose.

6.4 Substrate view

Each metabolic step is a substrate K-type rearrangement, catalyzed by an enzyme that provides a substrate-K-type-optimal saddle.

The network architecture of metabolism (highly conserved across organisms) reflects substrate's natural pathway-architecture for biological chemistry.

6.5 Information ↔ catalysis

Genes (Vol 13 Ch 2) encode enzymes (this chapter). Enzymes catalyze reactions (chemistry). Reactions reproduce DNA. Bootstrap loop: information ↔ catalysis ↔ reproduction.

BST view: this loop IS substrate's biological-scale K-type autopoiesis (self-creation). Closes the substrate computational reading of life: substrate K-types do biology.

6.6 K-audit anchors

- Vol 12 Ch 5 (kinetics, transition states)
- Vol 6 (thermodynamics)
- Vol 14 (information theory)

Level 3 — 5th-grader accessibility

Biochemistry = chemistry of life. Enzymes speed up reactions by 10^5-10^{17}x. ATP: universal energy currency. Metabolism: glucose + $O_2 \rightarrow CO_2 + H_2O$ + ATP (~30 per glucose). In BST: each step is a substrate K-type rearrangement; enzymes provide optimal substrate paths. Information ↔ catalysis ↔ reproduction: substrate's biological autopoiesis loop.

What comes next

Chapter 7: Protein Structure (existing BST_Vol13 chapter).

Where to look this up

- Lehninger, *Principles of Biochemistry*
- BST: Vol 12 Ch 5; Vol 6; Vol 14